FALL 2024

CSE-4355 Electromechanical Systems and Sensors

THE UNIVERSITY OF TEXAS AT ARLINGTON



LAB 1 REPORT

Submitted toward the partial completion of the requirements for CSE 4355

By,

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Under the guidance of

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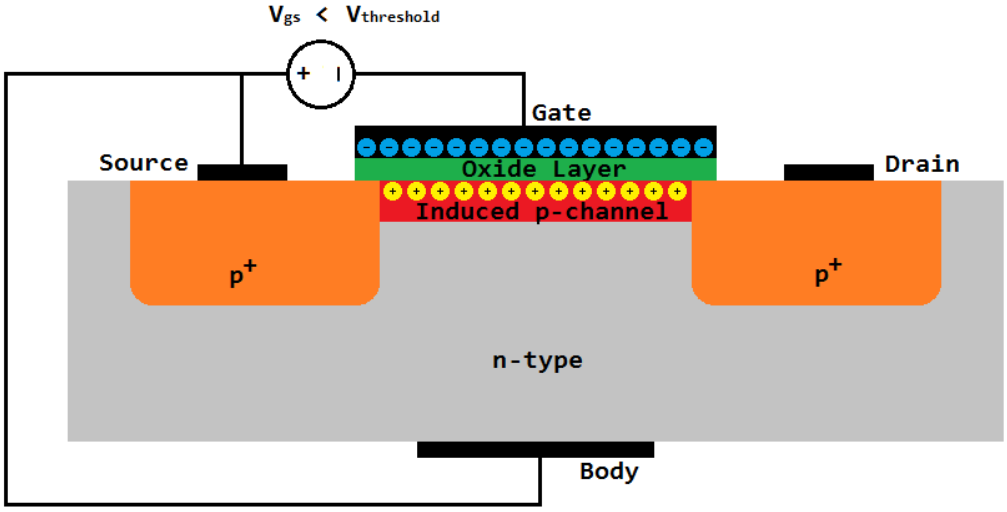
Lab Report:

Using Lab report 1 as an example, you should have all pictures taken, i.e. experimental setup, scope captures, and again any pictures of the breadboard should have its number clearly visible. You should have descriptive labels for any images, tables, or graphs included, you should have all measurements taken and explanations for them. For example, why did the large voltage spike occur, what did you measure it to be, what would this do to circuit components, what are some ways to mitigate it. This could then lead to why the flyback and Zener's were used, how it works based on the setup, what would happen with one Zener, two Zener's and so on. You can also include anything else you feel is related to the lab, that can help explain your observations, like discussing Lenz's law when explaining the direction of the back emf.

Introduction

Inductance is often overlooked in beginner circuits courses because of its unintuitive nature and seemingly less important attributes. However this constitutes a grave mistake when working with switching speeds and high frequencies, as the characteristics of an inductor cause high voltage spikes. Even though these spikes constitute an equivalent power to that of the running system, it creates excessive force (voltage) upon switching.

Background



This is an image of a typical mosfet design, with a gate, oxide layer, an induced channel, source, and drain. Mosfets are preferred for high speed switching as they operate based on voltage at the gate, with no current draw. This causes a problem when switching an inductive load such as a servo or solenoid because the counter EMF voltage spike can put a lot of pressure from the gate onto the oxide layer, and burst through it, destroying the part. The purpose of this lab is to explore induction and solutions that prevent such damage, while also maintaining switching speed.

Initial Experiments

Beginning the experiments, we first take a coil and apply a voltage to it, and then take the voltage away. A neon lamp with a striking voltage of 90 is used to observe which electrode lights up. Which electrode lighting up displayed the direction of current flow as the voltage feedback ran across the lamp, and it correlated with the polarity of the coil.

In the regular setup, the anode or resistor side of the dumet electrode glows on the neon tube. Reversing the power applied to the coil temporarily makes the cathode glow instead. Power applied in reverse also reverses the EMF voltage.

Afterwards, when using an oscilloscope probe, the Counter EMF voltage spike (Refer to background) seen from the coil during abrupt disconnection was -275V. Following, a measurement was taken using upper and lower limit switches on the coil to determine the time between mechanical completion of the coil charge. The times and voltages determined were as follows:

Spike to Upper limit = 40.4ms

Counter EMF Spike Amplitude > 305V

Spike to Lower limit = 101.6ms

Upper Limit to Lower limit = 61.2ms

And the Graph is shown here:

Diode placement and implementation

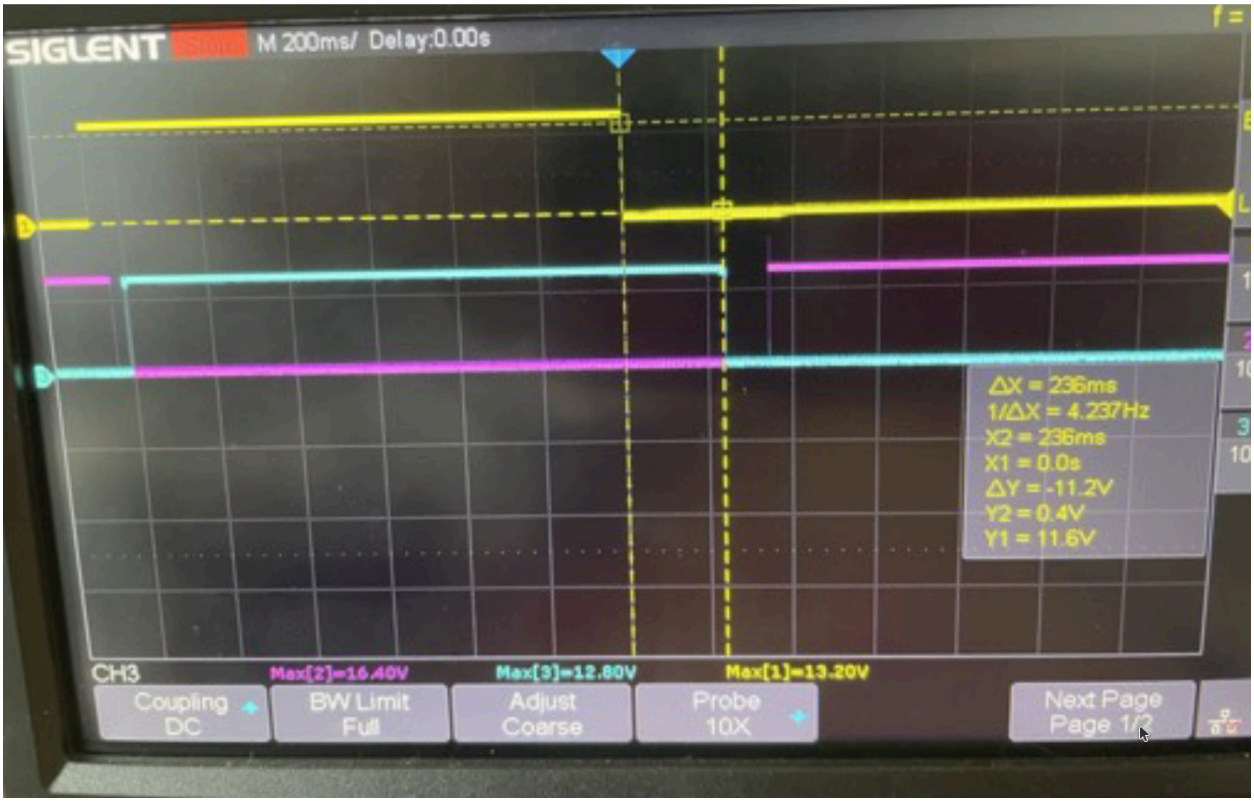
To remedy this high voltage, a diode is used as a dissipative force in the circuit. The same experiment as above was completed with a Schottky diode across the coil. The data and graph are again as follows:

With Schottky Diode,

Spike to Upper limit = 236 ms

Spike to Lower limit = 334ms

Upper Limit to Lower limit = 108ms



However, as a consequence of the voltage limit placed on the circuit from the diode (versus no limit and an excess of 300 volts) the coil takes much more time to dissipate its power. With high switching speeds this unfortunately slows down the coil.

As a solution, we can use higher voltage Zener Diodes for clamping, as they increase the voltage requirement to turn on the reverse flow and increase the voltage across the coil. This causes the coil dissipation time to drastically decrease. The same data run above is shown for Zener Diodes here:

Zener Diodes used:

-1N5913 3.3v

-1N5934 2.4v

Spike to upper limit = 45ms

Amp = 40

Spike to Lower limit = 105ms

Upper Limit to lower limit = 60ms



Results

From the above data, we can determine that Zener flyback diodes are the ideal diode for the circuit above. Using this knowledge, a new circuit is developed to show the highest achievable switching speed.

With the Schottky Diode:

Yellow = Low Side

Blue = High Side

Purple = Across 0.5

The magnetic field was on for 65,000 ms and off for 300,000 ms was the fastest parameters used while making sure it fully pressed both buttons.

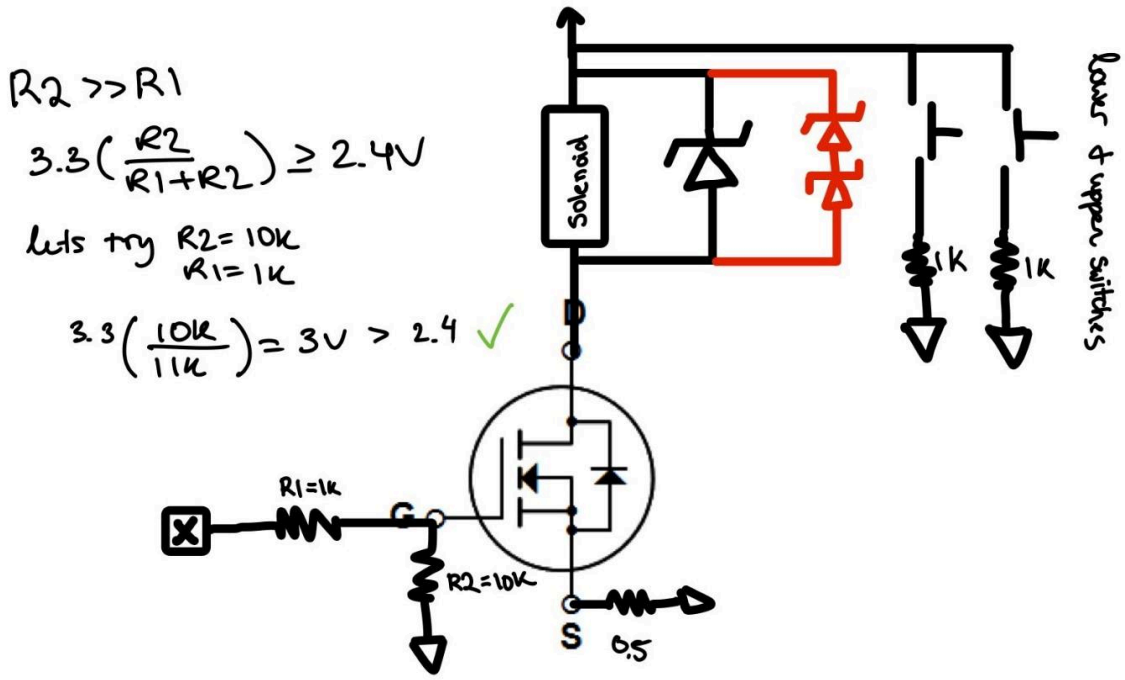
Zener and Schottky in series:

With the Zener diode(5934B) in series with the Schottky, we were able to lower the speed significantly, about 3 times faster.

Magnetic field on for 70,000 ms and off for 100,000 ms was the fastest parameters used while still fully pressing both buttons.



Circuit Diagram:



Final built circuit:

